

Sai Supriya Kottru

ALB + 2 EC2 (t3.micro) + EC2 consumer (t3.micro)

**Run 1 (BATCH\_SIZE=100, FLUSH\_INTERVAL=1000ms)**

153s

Infrastructure

Chat Server 1: 34.222.9.192 (t3.micro)

Chat Server 2: 52.38.251.140 (t3.micro)

RabbitMQ: 16.148.83.87 (t3.micro)

Consumer + Postgres: 34.209.85.116 (t3.micro)

ALB: [chatflow-alb-1032953489.us-west-2.elb.amazonaws.com](https://chatflow-alb-1032953489.us-west-2.elb.amazonaws.com)

Client (Test Load)

Pool: 128 connections

Rooms: 20

Users/room: 50

Messages/user: 500

Total messages: 500,000

Chat Server

PUBLISH\_WORKERS=20

ROOM\_COUNT=20

RABBIT\_HOST=172.31.25.198 (private IP)

Consumer-v3

CONSUMER\_WORKERS=5

DB\_WORKERS=3

STATS\_WORKERS=3

BATCH\_SIZE=100

FLUSH\_INTERVAL=1000ms

METRICS\_PORT=8080

DB: Postgres on 127.0.0.1:5432

RabbitMQ

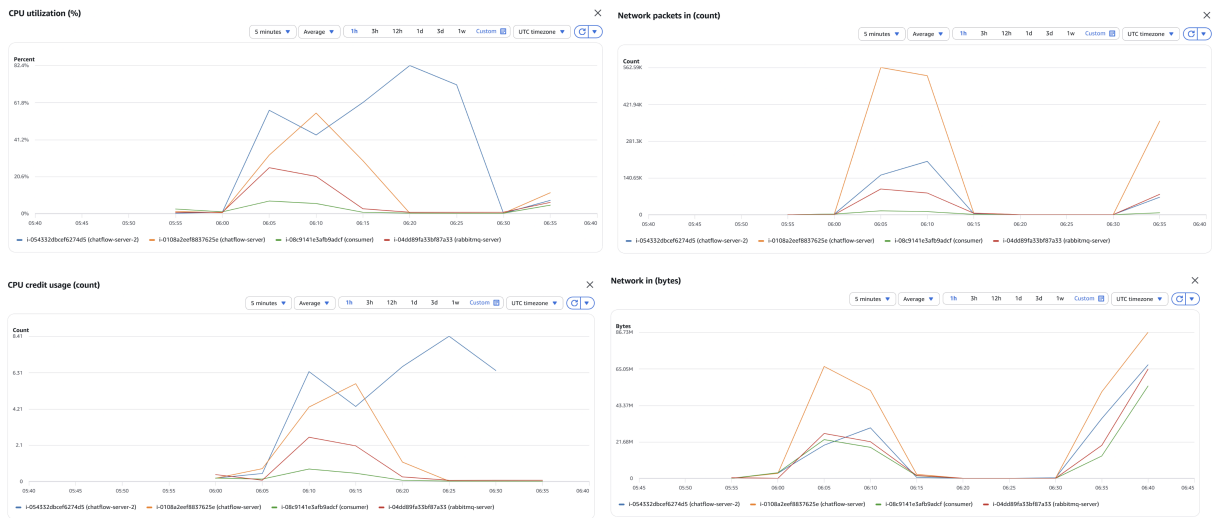
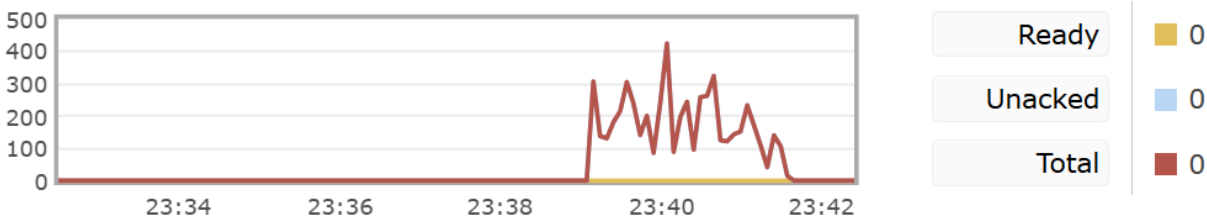
Exchange: chat.exchange (topic)

Persistence queue: persistence\_queue, binding room.#

Per-server room queues: 20 auto-delete queues per server

InitialCredits=500 per consumer

Queued messages last ten minutes ?



```
=== Performance Metrics ===
Total runtime (wall time): 153.3s
Successful messages sent: 500000
Failed messages: 0
Overall throughput: 3261.9 msg/s
Total connections: 1000
Reconnections: 24000
Failed connections: 0
Mean Latency across all rooms: 402ms
Median Latency across all rooms: 381ms
95th Percentile Latency across all rooms: 560ms
99th Percentile Latency across all rooms: 612ms
Min Latency across all rooms: 253ms
Max Latency across all rooms: 706ms
Median Throughput across all rooms: 718.19 msg/s
Throughput across rooms:

Room 17 | users: 50 | throughput: 35901.6 msg/s | mean latency: 502ms | median latency: 498ms
Message Types: JOIN: 36330 LEAVE: 35119 TEXT: 647783

Room 16 | users: 50 | throughput: 35792.6 msg/s | mean latency: 355ms | median latency: 356ms
Message Types: JOIN: 33831 TEXT: 608513 LEAVE: 33501

Room 3 | users: 50 | throughput: 32555.2 msg/s | mean latency: 344ms | median latency: 345ms
Message Types: JOIN: 29969 TEXT: 547182 LEAVE: 29993

Room 11 | users: 50 | throughput: 30805.1 msg/s | mean latency: 383ms | median latency: 382ms
Message Types: JOIN: 29904 TEXT: 535236 LEAVE: 28886

Room 9 | users: 50 | throughput: 18120.2 msg/s | mean latency: 483ms | median latency: 473ms
Message Types: JOIN: 18996 TEXT: 337867 LEAVE: 17966

Room 8 | users: 50 | throughput: 34143.7 msg/s | mean latency: 532ms | median latency: 537ms
Message Types: JOIN: 36542 TEXT: 655852 LEAVE: 36329

Room 5 | users: 50 | throughput: 38188.5 msg/s | mean latency: 337ms | median latency: 336ms
Message Types: TEXT: 534478 LEAVE: 29103 JOIN: 29909

Room 7 | users: 50 | throughput: 27266.0 msg/s | mean latency: 571ms | median latency: 564ms
Message Types: JOIN: 29814 LEAVE: 29308 TEXT: 533801
```

# Sai Supriya Kottru

```
Room 13 | users: 50 | throughput: 32650.0 msg/s | mean latency: 432ms | median latency: 434ms
Message Types: JOIN: 31574 TEXT: 561760 LEAVE: 31186

Room 10 | users: 50 | throughput: 39955.6 msg/s | mean latency: 411ms | median latency: 413ms
Message Types: LEAVE: 37492 JOIN: 37510 TEXT: 676990

Room 14 | users: 50 | throughput: 35336.3 msg/s | mean latency: 359ms | median latency: 363ms
Message Types: JOIN: 32634 TEXT: 592381 LEAVE: 32986

Room 15 | users: 50 | throughput: 37090.2 msg/s | mean latency: 338ms | median latency: 339ms
Message Types: JOIN: 35182 TEXT: 635134 LEAVE: 35132

Room 12 | users: 50 | throughput: 38460.8 msg/s | mean latency: 333ms | median latency: 334ms
Message Types: TEXT: 642362 LEAVE: 35417 JOIN: 35638

Room 1 | users: 50 | throughput: 32673.5 msg/s | mean latency: 401ms | median latency: 403ms
Message Types: LEAVE: 31394 JOIN: 32065 TEXT: 573295

Room 18 | users: 50 | throughput: 42510.4 msg/s | mean latency: 291ms | median latency: 291ms
Message Types: LEAVE: 26200 JOIN: 26253 TEXT: 473306

Room 19 | users: 50 | throughput: 36598.1 msg/s | mean latency: 501ms | median latency: 498ms
Message Types: TEXT: 659358 LEAVE: 35581 JOIN: 37043

Room 20 | users: 50 | throughput: 40241.8 msg/s | mean latency: 332ms | median latency: 333ms
Message Types: TEXT: 698191 LEAVE: 38781 JOIN: 38782

Room 2 | users: 50 | throughput: 39737.6 msg/s | mean latency: 328ms | median latency: 329ms
Message Types: LEAVE: 37324 JOIN: 37247 TEXT: 675108

Room 4 | users: 50 | throughput: 38691.8 msg/s | mean latency: 405ms | median latency: 404ms
Message Types: LEAVE: 37843 JOIN: 38165 TEXT: 683544

Room 6 | users: 50 | throughput: 23895.4 msg/s | mean latency: 404ms | median latency: 437ms
Message Types: JOIN: 23505 TEXT: 422257 LEAVE: 23344

===== Consumer Metrics =====

--- Core ---
Active users (last 1h): 50
Busiest room: 9 (44330 messages)
Most active user: 24 (20 rooms)
  room 18 last active: 2026-03-25T06:41:31.454454Z
  room 5 last active: 2026-03-25T06:41:27.231932Z
  room 16 last active: 2026-03-25T06:41:19.013496Z
  room 4 last active: 2026-03-25T06:41:14.143904Z
  room 1 last active: 2026-03-25T06:41:07.679433Z
  room 12 last active: 2026-03-25T06:40:55.905761Z
  room 15 last active: 2026-03-25T06:40:53.004391Z
  room 14 last active: 2026-03-25T06:40:41.608304Z
  room 10 last active: 2026-03-25T06:40:37.842042Z
  room 13 last active: 2026-03-25T06:40:31.369464Z
  room 8 last active: 2026-03-25T06:40:22.675081Z
  room 7 last active: 2026-03-25T06:40:17.835428Z
  room 9 last active: 2026-03-25T06:40:00.585227Z
  room 11 last active: 2026-03-25T06:39:55.058757Z
  room 3 last active: 2026-03-25T06:39:53.599825Z
  room 2 last active: 2026-03-25T06:39:40.470823Z
  room 20 last active: 2026-03-25T06:39:38.094066Z
  room 6 last active: 2026-03-25T06:39:30.603702Z
  room 17 last active: 2026-03-25T06:39:20.686583Z
  room 19 last active: 2026-03-25T06:39:15.849211Z

--- Messages/min (last 15 buckets) ---
2026-03-25T06:41:00Z 104668 msg
2026-03-25T06:40:00Z 193546 msg
2026-03-25T06:39:00Z 196072 msg
2026-03-25T06:38:00Z 5714 msg
2026-03-25T06:17:00Z 11100 msg
2026-03-25T06:13:00Z 31917 msg
2026-03-25T06:12:00Z 35161 msg
2026-03-25T06:11:00Z 33955 msg
2026-03-25T06:10:00Z 33283 msg
2026-03-25T06:09:00Z 21937 msg
2026-03-25T06:08:00Z 20413 msg
2026-03-25T06:07:00Z 33541 msg
2026-03-25T06:06:00Z 23806 msg
2026-03-25T06:05:00Z 67937 msg
```

```
--- Top 5 users by message count ---
1. 24 - 17151 messages
2. 35 - 16989 messages
3. 49 - 16906 messages
4. 12 - 16898 messages
5. 5 - 16749 messages

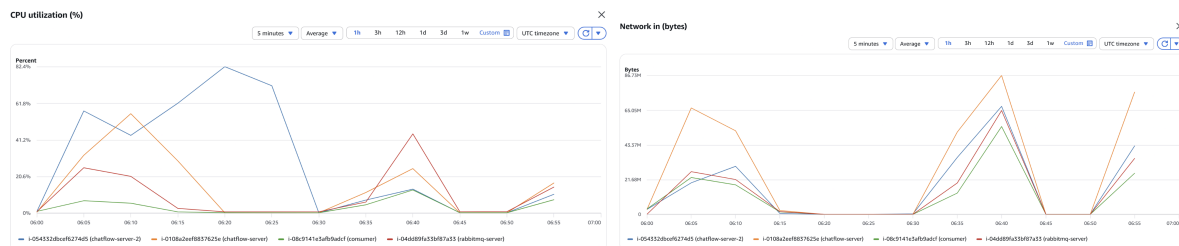
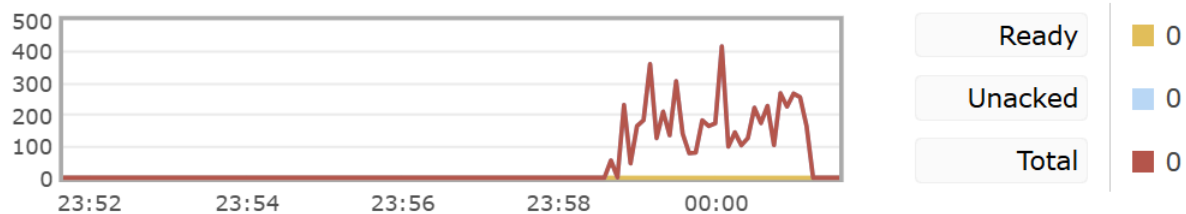
--- Top 5 rooms by message count ---
1. 9 - 44330 messages
2. 10 - 41594 messages
3. 16 - 41513 messages
4. 19 - 41301 messages
5. 17 - 41088 messages

--- Top 5 users by room count ---
1. 8 - 20 rooms
2. 12 - 20 rooms
3. 10 - 20 rooms
4. 7 - 20 rooms
5. 49 - 20 rooms
=====
```

**BATCH\_SIZE=500, FLUSH\_INTERVAL=500ms**

161.0s

Queued messages last ten minutes ?





Sai Supriya Kotturu



```
=== Performance Metrics ===
Total runtime (wall time): 161.0s
Successful messages sent: 500000
Failed messages: 0
Overall throughput: 3106.4 msg/s
Total connections: 1000
Reconnections: 24000
Failed connections: 0
Mean Latency across all rooms: 421ms
Median Latency across all rooms: 412ms
95th Percentile Latency across all rooms: 585ms
99th Percentile Latency across all rooms: 648ms
Min Latency across all rooms: 303ms
Max Latency across all rooms: 852ms
Median Throughput across all rooms: 701.45 msg/s
Throughput across rooms:

Room 19 | users: 50 | throughput: 20714.8 msg/s | mean latency: 434ms | median latency: 431ms
Message Types: TEXT: 362323 LEAVE: 19946 JOIN: 20106

Room 16 | users: 50 | throughput: 31974.0 msg/s | mean latency: 470ms | median latency: 469ms
Message Types: TEXT: 608797 LEAVE: 31931 JOIN: 35111

Room 15 | users: 50 | throughput: 40492.7 msg/s | mean latency: 475ms | median latency: 475ms
Message Types: TEXT: 716468 LEAVE: 39359 JOIN: 40371

Room 3 | users: 50 | throughput: 39696.6 msg/s | mean latency: 341ms | median latency: 343ms
Message Types: TEXT: 668100 JOIN: 36551 LEAVE: 36710

Room 6 | users: 50 | throughput: 39445.5 msg/s | mean latency: 345ms | median latency: 346ms
Message Types: LEAVE: 37114 JOIN: 37004 TEXT: 673748

Room 13 | users: 50 | throughput: 28122.8 msg/s | mean latency: 429ms | median latency: 455ms
Message Types: LEAVE: 31352 JOIN: 31166 TEXT: 568181

Room 1 | users: 50 | throughput: 35818.9 msg/s | mean latency: 492ms | median latency: 493ms
Message Types: JOIN: 32669 TEXT: 577043 LEAVE: 31310

Room 18 | users: 50 | throughput: 40084.3 msg/s | mean latency: 337ms | median latency: 336ms
Message Types: LEAVE: 37683 JOIN: 37675 TEXT: 679497
```

```
Room 7 | users: 50 | throughput: 36460.4 msg/s | mean latency: 364ms | median latency: 361ms
Message Types: TEXT: 647763 LEAVE: 35736 JOIN: 35578

Room 4 | users: 50 | throughput: 36604.2 msg/s | mean latency: 356ms | median latency: 357ms
Message Types: LEAVE: 35211 JOIN: 35175 TEXT: 635698

Room 20 | users: 50 | throughput: 34430.9 msg/s | mean latency: 426ms | median latency: 429ms
Message Types: LEAVE: 33194 JOIN: 33980 TEXT: 609525

Room 9 | users: 50 | throughput: 34401.8 msg/s | mean latency: 452ms | median latency: 449ms
Message Types: TEXT: 440600 LEAVE: 24227 JOIN: 24688

Room 2 | users: 50 | throughput: 27613.6 msg/s | mean latency: 601ms | median latency: 598ms
Message Types: LEAVE: 35553 JOIN: 35311 TEXT: 642876

Room 8 | users: 50 | throughput: 34014.8 msg/s | mean latency: 352ms | median latency: 353ms
Message Types: JOIN: 33803 TEXT: 612786 LEAVE: 33992

Room 14 | users: 50 | throughput: 31608.4 msg/s | mean latency: 464ms | median latency: 464ms
Message Types: LEAVE: 30871 JOIN: 34509 TEXT: 595160

Room 12 | users: 50 | throughput: 39778.7 msg/s | mean latency: 467ms | median latency: 468ms
Message Types: JOIN: 40496 TEXT: 724032 LEAVE: 39446

Room 11 | users: 50 | throughput: 32775.4 msg/s | mean latency: 339ms | median latency: 338ms
Message Types: JOIN: 30172 TEXT: 543936 LEAVE: 30124

Room 10 | users: 50 | throughput: 30343.8 msg/s | mean latency: 336ms | median latency: 335ms
Message Types: JOIN: 28273 TEXT: 510728 LEAVE: 28302

Room 5 | users: 50 | throughput: 30711.0 msg/s | mean latency: 578ms | median latency: 582ms
Message Types: JOIN: 39382 TEXT: 714797 LEAVE: 39519

Room 17 | users: 50 | throughput: 36066.0 msg/s | mean latency: 367ms | median latency: 368ms
Message Types: LEAVE: 34849 JOIN: 34937 TEXT: 630102
```

```
===== Consumer Metrics =====
```

```
--- Core ---
Active users (last 1h): 50
Busiest room: 10 (25000 messages)
Most active user: 21 (20 rooms)
  room 9      last active: 2026-03-25T07:01:10.079704Z
  room 1      last active: 2026-03-25T07:01:07.560638Z
  room 20     last active: 2026-03-25T07:00:55.277006Z
  room 17     last active: 2026-03-25T07:00:49.958927Z
  room 8      last active: 2026-03-25T07:00:43.443482Z
  room 4      last active: 2026-03-25T07:00:34.215544Z
  room 7      last active: 2026-03-25T07:00:29.025981Z
  room 2      last active: 2026-03-25T07:00:22.475388Z
  room 5      last active: 2026-03-25T07:00:16.171583Z
  room 13     last active: 2026-03-25T07:00:00.416634Z
  room 6      last active: 2026-03-25T06:59:52.214767Z
  room 18     last active: 2026-03-25T06:59:43.725502Z
  room 10     last active: 2026-03-25T06:59:35.941714Z
  room 3      last active: 2026-03-25T06:59:32.791516Z
  room 11     last active: 2026-03-25T06:59:25.654214Z
  room 12     last active: 2026-03-25T06:59:13.012371Z
  room 15     last active: 2026-03-25T06:59:08.457241Z
  room 16     last active: 2026-03-25T06:58:52.727144Z
  room 14     last active: 2026-03-25T06:58:52.72553Z
  room 19     last active: 2026-03-25T06:58:51.356713Z

--- Messages/min (last 15 buckets) ---
2026-03-25T07:01:00Z  34202 msg
2026-03-25T07:00:00Z  174663 msg
2026-03-25T06:59:00Z  201321 msg
2026-03-25T06:58:00Z  89794 msg
```

```
--- Top 5 users by message count ---
1. 25 - 10000 messages
2. 40 - 10000 messages
3. 44 - 10000 messages
4. 8 - 10000 messages
5. 21 - 10000 messages

--- Top 5 rooms by message count ---
1. 7 - 25000 messages
2. 20 - 25000 messages
3. 5 - 25000 messages
4. 3 - 25000 messages
5. 10 - 25000 messages

--- Top 5 users by room count ---
1. 8 - 20 rooms
2. 12 - 20 rooms
3. 10 - 20 rooms
4. 7 - 20 rooms
5. 49 - 20 rooms
```

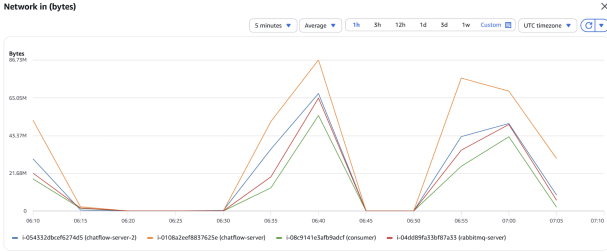
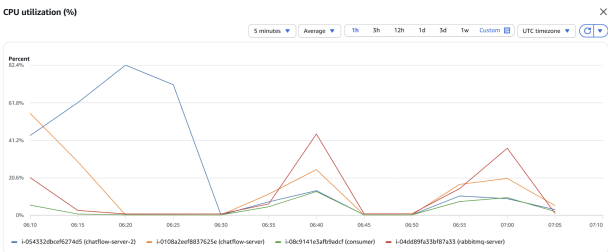
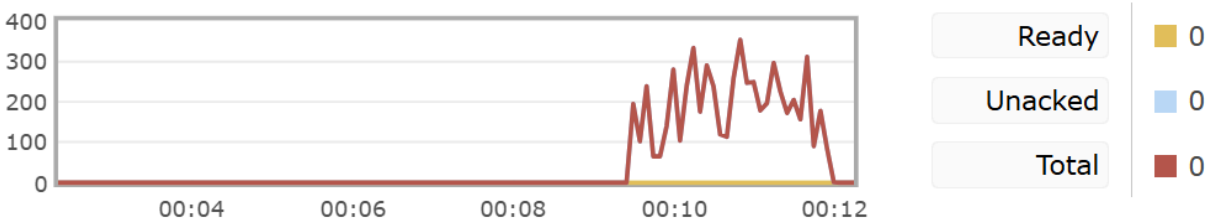
=====

→ chat-flow git:(assignment-3) 🌸 🍄 🍷 📺

BATCH\_SIZE=1000, FLUSH\_INTERVAL=500ms

156.4s

Queued messages last ten minutes ?



# Sai Supriya Kottru

```
=== Performance Metrics ===
Total runtime (wall time): 156.4s
Successful messages sent: 500000
Failed messages: 0
Overall throughput: 3196.1 msg/s
Total connections: 1000
Reconnections: 24000
Failed connections: 0
Mean Latency across all rooms: 393ms
Median Latency across all rooms: 381ms
95th Percentile Latency across all rooms: 487ms
99th Percentile Latency across all rooms: 514ms
Min Latency across all rooms: 313ms
Max Latency across all rooms: 539ms
Median Throughput across all rooms: 726.51 msg/s
Throughput across rooms:

Room 6 | users: 50 | throughput: 39236.9 msg/s | mean latency: 420ms | median latency: 424ms
Message Types: JOIN: 36367 TEXT: 645761 LEAVE: 34695

Room 3 | users: 50 | throughput: 24100.5 msg/s | mean latency: 371ms | median latency: 371ms
Message Types: JOIN: 22335 TEXT: 407904 LEAVE: 22265

Room 12 | users: 50 | throughput: 35620.9 msg/s | mean latency: 500ms | median latency: 502ms
Message Types: TEXT: 649926 LEAVE: 35921 JOIN: 36327

Room 11 | users: 50 | throughput: 36587.9 msg/s | mean latency: 385ms | median latency: 383ms
Message Types: JOIN: 37334 TEXT: 676749 LEAVE: 37476

Room 18 | users: 50 | throughput: 33653.5 msg/s | mean latency: 355ms | median latency: 353ms
Message Types: JOIN: 33181 TEXT: 595463 LEAVE: 33215

Room 20 | users: 50 | throughput: 28103.8 msg/s | mean latency: 380ms | median latency: 378ms
Message Types: TEXT: 381499 LEAVE: 20752 JOIN: 21638

Room 19 | users: 50 | throughput: 39131.0 msg/s | mean latency: 423ms | median latency: 424ms
Message Types: LEAVE: 34958 JOIN: 36600 TEXT: 651319

Room 7 | users: 50 | throughput: 36400.4 msg/s | mean latency: 382ms | median latency: 393ms
Message Types: JOIN: 36993 TEXT: 672816 LEAVE: 37171

Room 8 | users: 50 | throughput: 31171.1 msg/s | mean latency: 446ms | median latency: 441ms
Message Types: TEXT: 614208 LEAVE: 34075 JOIN: 33956
```

```
Room 17 | users: 50 | throughput: 33942.8 msg/s | mean latency: 353ms | median latency: 352ms
Message Types: JOIN: 30552 TEXT: 559158 LEAVE: 30760

Room 10 | users: 50 | throughput: 37765.1 msg/s | mean latency: 363ms | median latency: 364ms
Message Types: JOIN: 34855 TEXT: 630883 LEAVE: 34703

Room 9 | users: 50 | throughput: 38032.4 msg/s | mean latency: 465ms | median latency: 465ms
Message Types: JOIN: 37104 TEXT: 649996 LEAVE: 35341

Room 13 | users: 50 | throughput: 37991.9 msg/s | mean latency: 346ms | median latency: 348ms
Message Types: LEAVE: 35592 JOIN: 35502 TEXT: 643517

Room 4 | users: 50 | throughput: 37028.2 msg/s | mean latency: 377ms | median latency: 376ms
Message Types: JOIN: 36842 TEXT: 665387 LEAVE: 36995

Room 16 | users: 50 | throughput: 31391.8 msg/s | mean latency: 428ms | median latency: 432ms
Message Types: LEAVE: 28414 JOIN: 29810 TEXT: 525223

Room 14 | users: 50 | throughput: 38853.9 msg/s | mean latency: 351ms | median latency: 351ms
Message Types: LEAVE: 38569 JOIN: 38541 TEXT: 697537

Room 1 | users: 50 | throughput: 39526.7 msg/s | mean latency: 349ms | median latency: 351ms
Message Types: JOIN: 36948 TEXT: 669774 LEAVE: 37031

Room 2 | users: 50 | throughput: 35986.6 msg/s | mean latency: 445ms | median latency: 443ms
Message Types: JOIN: 36742 TEXT: 668948 LEAVE: 36943

Room 15 | users: 50 | throughput: 36053.0 msg/s | mean latency: 357ms | median latency: 357ms
Message Types: TEXT: 610229 LEAVE: 33578 JOIN: 33735

Room 5 | users: 50 | throughput: 34116.7 msg/s | mean latency: 369ms | median latency: 371ms
Message Types: JOIN: 28675 TEXT: 517147 LEAVE: 27901

===== Consumer Metrics =====
```

```
===== Consumer Metrics =====  
  
--- Core ---  
Active users (last 1h): 50  
Busiest room:          16 (25000 messages)  
Most active user:      34 (20 rooms)  
  room 20                last active: 2026-03-25T07:11:56.880384Z  
  room 5                  last active: 2026-03-25T07:11:52.16216Z  
  room 15                 last active: 2026-03-25T07:11:47.436228Z  
  room 13                 last active: 2026-03-25T07:11:38.668876Z  
  room 18                 last active: 2026-03-25T07:11:30.030247Z  
  room 11                 last active: 2026-03-25T07:11:25.063256Z  
  room 2                  last active: 2026-03-25T07:11:18.439285Z  
  room 12                 last active: 2026-03-25T07:11:06.650134Z  
  room 9                  last active: 2026-03-25T07:11:00.068654Z  
  room 16                 last active: 2026-03-25T07:10:53.664242Z  
  room 10                 last active: 2026-03-25T07:10:43.725892Z  
  room 1                  last active: 2026-03-25T07:10:40.880867Z  
  room 17                 last active: 2026-03-25T07:10:25.746908Z  
  room 4                  last active: 2026-03-25T07:10:22.808835Z  
  room 8                  last active: 2026-03-25T07:10:16.14785Z  
  room 7                  last active: 2026-03-25T07:10:09.125991Z  
  room 14                 last active: 2026-03-25T07:09:57.832437Z  
  room 6                  last active: 2026-03-25T07:09:44.093018Z  
  room 19                 last active: 2026-03-25T07:09:42.193496Z  
  room 3                  last active: 2026-03-25T07:09:41.578901Z
```

```
--- Messages/min (last 15 buckets) ---  
2026-03-25T07:11:00Z 181326 msg  
2026-03-25T07:10:00Z 195160 msg  
2026-03-25T07:09:00Z 122879 msg
```

```
--- Top 5 users by message count ---  
1. 34 - 9997 messages  
2. 21 - 9995 messages  
3. 16 - 9995 messages  
4. 45 - 9994 messages  
5. 33 - 9994 messages
```

```
--- Top 5 rooms by message count ---  
1. 17 - 25000 messages  
2. 16 - 25000 messages  
3. 14 - 25000 messages  
4. 8 - 25000 messages  
5. 10 - 25000 messages
```

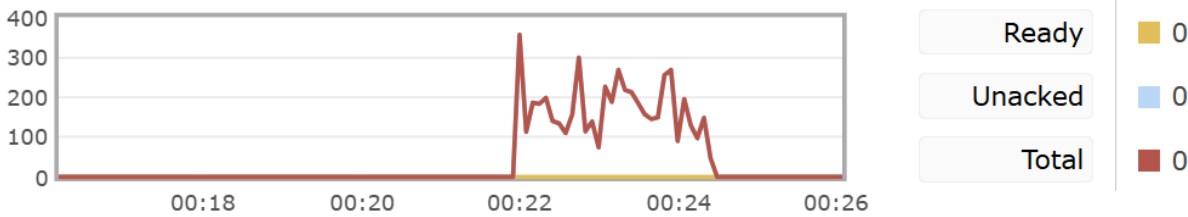
```
--- Top 5 users by room count ---  
1. 8 - 20 rooms  
2. 12 - 20 rooms  
3. 10 - 20 rooms  
4. 7 - 20 rooms  
5. 49 - 20 rooms
```

```
=====
```

**BATCH\_SIZE=5000, FLUSH\_INTERVAL=100ms**

154.7s

Queued messages last ten minutes ?



CPU utilization (%)



Network in (Bytes)



Network packets in (count)



CPU credit usage (count)



# Sai Supriya Kottru

```
=== Performance Metrics ===
Total runtime (wall time): 154.7s
Successful messages sent: 500000
Failed messages: 0
Overall throughput: 3232.0 msg/s
Total connections: 1000
Reconnections: 24000
Failed connections: 0
Mean Latency across all rooms: 368ms
Median Latency across all rooms: 359ms
95th Percentile Latency across all rooms: 463ms
99th Percentile Latency across all rooms: 488ms
Min Latency across all rooms: 200ms
Max Latency across all rooms: 522ms
Median Throughput across all rooms: 741.87 msg/s
Throughput across rooms:

Room 11 | users: 50 | throughput: 38920.5 msg/s | mean latency: 456ms | median latency: 456ms
Message Types: LEAVE: 36388 JOIN: 37693 TEXT: 672481

Room 13 | users: 50 | throughput: 33197.3 msg/s | mean latency: 340ms | median latency: 342ms
Message Types: JOIN: 31315 TEXT: 566091 LEAVE: 31479

Room 18 | users: 50 | throughput: 42357.1 msg/s | mean latency: 235ms | median latency: 237ms
Message Types: LEAVE: 25059 JOIN: 24949 TEXT: 448628

Room 5 | users: 50 | throughput: 37736.7 msg/s | mean latency: 457ms | median latency: 459ms
Message Types: TEXT: 680355 LEAVE: 36843 JOIN: 38179

Room 9 | users: 50 | throughput: 39019.4 msg/s | mean latency: 360ms | median latency: 360ms
Message Types: JOIN: 38160 TEXT: 690698 LEAVE: 38272

Room 19 | users: 50 | throughput: 34074.5 msg/s | mean latency: 349ms | median latency: 349ms
Message Types: TEXT: 601947 LEAVE: 33450 JOIN: 33327

Room 4 | users: 50 | throughput: 39612.4 msg/s | mean latency: 343ms | median latency: 345ms
Message Types: TEXT: 674149 LEAVE: 37305 JOIN: 37097

Room 12 | users: 50 | throughput: 34012.5 msg/s | mean latency: 368ms | median latency: 362ms
Message Types: JOIN: 33846 TEXT: 609380 LEAVE: 33888

Room 10 | users: 50 | throughput: 34381.8 msg/s | mean latency: 430ms | median latency: 428ms
Message Types: JOIN: 35154 TEXT: 640352 LEAVE: 35397
```

```
Room 10 | users: 50 | throughput: 34381.8 msg/s | mean latency: 430ms | median latency: 428ms
Message Types: JOIN: 35154 TEXT: 640352 LEAVE: 35397

Room 14 | users: 50 | throughput: 38871.6 msg/s | mean latency: 326ms | median latency: 321ms
Message Types: LEAVE: 33572 JOIN: 33512 TEXT: 603251

Room 6 | users: 50 | throughput: 24549.5 msg/s | mean latency: 387ms | median latency: 403ms
Message Types: LEAVE: 23951 JOIN: 23986 TEXT: 432661

Room 1 | users: 50 | throughput: 39149.3 msg/s | mean latency: 358ms | median latency: 354ms
Message Types: JOIN: 36994 TEXT: 669236 LEAVE: 37027

Room 2 | users: 50 | throughput: 33942.6 msg/s | mean latency: 357ms | median latency: 357ms
Message Types: JOIN: 32849 TEXT: 598945 LEAVE: 32954

Room 7 | users: 50 | throughput: 36811.2 msg/s | mean latency: 355ms | median latency: 356ms
Message Types: TEXT: 634882 LEAVE: 35322 JOIN: 35159

Room 15 | users: 50 | throughput: 36867.2 msg/s | mean latency: 358ms | median latency: 359ms
Message Types: TEXT: 652992 LEAVE: 36217 JOIN: 36177

Room 8 | users: 50 | throughput: 33748.9 msg/s | mean latency: 447ms | median latency: 450ms
Message Types: JOIN: 35958 TEXT: 651803 LEAVE: 36009

Room 17 | users: 50 | throughput: 38779.9 msg/s | mean latency: 363ms | median latency: 363ms
Message Types: TEXT: 685228 LEAVE: 37935 JOIN: 37851

Room 16 | users: 50 | throughput: 37613.0 msg/s | mean latency: 368ms | median latency: 369ms
Message Types: JOIN: 37257 TEXT: 670387 LEAVE: 37346

Room 20 | users: 50 | throughput: 37970.0 msg/s | mean latency: 358ms | median latency: 360ms
Message Types: JOIN: 37132 TEXT: 669480 LEAVE: 37258

Room 3 | users: 50 | throughput: 36077.5 msg/s | mean latency: 354ms | median latency: 355ms
Message Types: JOIN: 34744 TEXT: 630552 LEAVE: 34828
```

```
===== Consumer Metrics =====
```

=====  
Consumer Metrics  
=====

--- Core ---

Active users (last 1h): 50

Busiest room: 6 (25000 messages)

Most active user: 19 (20 rooms)

|         |                                          |
|---------|------------------------------------------|
| room 18 | last active: 2026-03-25T07:24:25.156369Z |
| room 14 | last active: 2026-03-25T07:24:19.606578Z |
| room 10 | last active: 2026-03-25T07:24:16.075387Z |
| room 8  | last active: 2026-03-25T07:24:10.651957Z |
| room 12 | last active: 2026-03-25T07:23:58.172464Z |
| room 15 | last active: 2026-03-25T07:23:54.0796Z   |
| room 3  | last active: 2026-03-25T07:23:53.103651Z |
| room 7  | last active: 2026-03-25T07:23:38.787828Z |
| room 20 | last active: 2026-03-25T07:23:31.121639Z |
| room 2  | last active: 2026-03-25T07:23:27.584532Z |
| room 16 | last active: 2026-03-25T07:23:14.381836Z |
| room 17 | last active: 2026-03-25T07:23:04.771956Z |
| room 13 | last active: 2026-03-25T07:22:56.357127Z |
| room 4  | last active: 2026-03-25T07:22:49.770837Z |
| room 19 | last active: 2026-03-25T07:22:46.471216Z |
| room 1  | last active: 2026-03-25T07:22:30.719316Z |
| room 9  | last active: 2026-03-25T07:22:30.581213Z |
| room 11 | last active: 2026-03-25T07:22:17.077756Z |
| room 6  | last active: 2026-03-25T07:22:10.409483Z |
| room 5  | last active: 2026-03-25T07:22:10.409146Z |

--- Messages/min (last 15 buckets) ---

|                      |            |
|----------------------|------------|
| 2026-03-25T07:24:00Z | 82466 msg  |
| 2026-03-25T07:23:00Z | 196082 msg |
| 2026-03-25T07:22:00Z | 195312 msg |
| 2026-03-25T07:21:00Z | 26024 msg  |

--- Top 5 users by message count ---

1. 17 - 10000 messages
2. 33 - 10000 messages
3. 19 - 10000 messages
4. 2 - 10000 messages
5. 38 - 10000 messages

--- Top 5 rooms by message count ---

1. 3 - 25000 messages
2. 6 - 25000 messages
3. 1 - 25000 messages
4. 9 - 25000 messages
5. 11 - 25000 messages

--- Top 5 users by room count ---

1. 8 - 20 rooms
2. 12 - 20 rooms
3. 10 - 20 rooms
4. 7 - 20 rooms
5. 49 - 20 rooms

=====

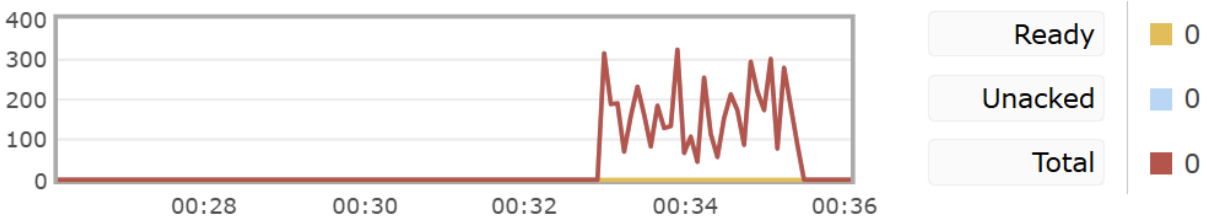
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BATCH\_SIZE=5000, FLUSH\_INTERVAL=1000ms

152.7s

Queued messages last ten minutes ?



```
=== Performance Metrics ===
Total runtime (wall time): 152.7s
Successful messages sent: 500000
Failed messages: 0
Overall throughput: 3273.9 msg/s
Total connections: 1000
Reconnections: 24000
Failed connections: 0
Mean Latency across all rooms: 379ms
Median Latency across all rooms: 374ms
95th Percentile Latency across all rooms: 462ms
99th Percentile Latency across all rooms: 485ms
Min Latency across all rooms: 214ms
Max Latency across all rooms: 502ms
Median Throughput across all rooms: 748.54 msg/s
Throughput across rooms:

Room 9 | users: 50 | throughput: 37088.2 msg/s | mean latency: 387ms | median latency: 394ms
Message Types: JOIN: 36943 TEXT: 668078 LEAVE: 37040

Room 17 | users: 50 | throughput: 33655.0 msg/s | mean latency: 415ms | median latency: 417ms
Message Types: TEXT: 600659 LEAVE: 33319 JOIN: 33269

Room 18 | users: 50 | throughput: 38985.4 msg/s | mean latency: 354ms | median latency: 358ms
Message Types: TEXT: 667129 LEAVE: 36808 JOIN: 36527

Room 14 | users: 50 | throughput: 40173.7 msg/s | mean latency: 348ms | median latency: 348ms
Message Types: TEXT: 702414 LEAVE: 38809 JOIN: 38673

Room 11 | users: 50 | throughput: 39085.2 msg/s | mean latency: 361ms | median latency: 362ms
Message Types: LEAVE: 36836 JOIN: 36522 TEXT: 668459

Room 7 | users: 50 | throughput: 34083.5 msg/s | mean latency: 347ms | median latency: 345ms
Message Types: JOIN: 31874 TEXT: 576202 LEAVE: 31937

Room 10 | users: 50 | throughput: 36415.7 msg/s | mean latency: 430ms | median latency: 431ms
Message Types: LEAVE: 36145 JOIN: 35940 TEXT: 658312

Room 2 | users: 50 | throughput: 38783.8 msg/s | mean latency: 350ms | median latency: 349ms
Message Types: TEXT: 600085 LEAVE: 36298 JOIN: 36205

Room 16 | users: 50 | throughput: 35193.8 msg/s | mean latency: 441ms | median latency: 441ms
Message Types: TEXT: 652699 LEAVE: 36147 JOIN: 35804
```

```
Room 8 | users: 50 | throughput: 24513.9 msg/s | mean latency: 382ms | median latency: 393ms
Message Types: TEXT: 420940 JOIN: 23291 LEAVE: 23088

Room 15 | users: 50 | throughput: 39855.4 msg/s | mean latency: 353ms | median latency: 356ms
Message Types: LEAVE: 37270 JOIN: 37088 TEXT: 678719

Room 19 | users: 50 | throughput: 36695.9 msg/s | mean latency: 359ms | median latency: 362ms
Message Types: LEAVE: 35718 JOIN: 35620 TEXT: 644523

Room 20 | users: 50 | throughput: 35540.5 msg/s | mean latency: 386ms | median latency: 387ms
Message Types: LEAVE: 33080 JOIN: 32804 TEXT: 603498

Room 13 | users: 50 | throughput: 37063.5 msg/s | mean latency: 391ms | median latency: 391ms
Message Types: JOIN: 36022 TEXT: 656965 LEAVE: 36163

Room 1 | users: 50 | throughput: 44813.2 msg/s | mean latency: 236ms | median latency: 234ms
Message Types: JOIN: 28471 LEAVE: 28479 TEXT: 513066

Room 4 | users: 50 | throughput: 34224.8 msg/s | mean latency: 356ms | median latency: 357ms
Message Types: LEAVE: 31471 JOIN: 31286 TEXT: 571052

Room 12 | users: 50 | throughput: 37419.1 msg/s | mean latency: 370ms | median latency: 372ms
Message Types: LEAVE: 34884 JOIN: 34690 TEXT: 630566

Room 6 | users: 50 | throughput: 37135.8 msg/s | mean latency: 409ms | median latency: 411ms
Message Types: LEAVE: 33502 JOIN: 33193 TEXT: 607924

Room 5 | users: 50 | throughput: 37649.5 msg/s | mean latency: 449ms | median latency: 449ms
Message Types: LEAVE: 34992 JOIN: 36941 TEXT: 654204

Room 3 | users: 50 | throughput: 38000.2 msg/s | mean latency: 453ms | median latency: 454ms
Message Types: TEXT: 681677 LEAVE: 36245 JOIN: 38623

===== Consumer Metrics =====
```

```
--- Core ---
Active users (last 1h): 50
Busiest room:          13 (25000 messages)
Most active user:      39 (20 rooms)
  room 1                last active: 2026-03-25T07:35:18.687749Z
  room 6                last active: 2026-03-25T07:35:15.264347Z
  room 16               last active: 2026-03-25T07:35:04.690407Z
  room 13               last active: 2026-03-25T07:34:59.138897Z
  room 20               last active: 2026-03-25T07:34:47.826457Z
  room 18               last active: 2026-03-25T07:34:40.174925Z
  room 19               last active: 2026-03-25T07:34:35.181111Z
  room 12               last active: 2026-03-25T07:34:30.045454Z
  room 2                last active: 2026-03-25T07:34:20.989464Z
  room 10               last active: 2026-03-25T07:34:08.066565Z
  room 17               last active: 2026-03-25T07:34:06.723341Z
  room 9                last active: 2026-03-25T07:34:06.000832Z
  room 7                last active: 2026-03-25T07:33:51.19318Z
  room 11               last active: 2026-03-25T07:33:41.50871Z
  room 4                last active: 2026-03-25T07:33:30.466422Z
  room 14               last active: 2026-03-25T07:33:25.930452Z
  room 15               last active: 2026-03-25T07:33:25.858004Z
  room 3                last active: 2026-03-25T07:33:09.817484Z
  room 8                last active: 2026-03-25T07:33:04.341719Z
  room 5                last active: 2026-03-25T07:33:04.13381Z
```

```
--- Messages/min (last 15 buckets) ---
2026-03-25T07:35:00Z 61109 msg
2026-03-25T07:34:00Z 197594 msg
2026-03-25T07:33:00Z 198718 msg
2026-03-25T07:32:00Z 42579 msg
```

```
--- Top 5 users by message count ---
1. 44 - 10000 messages
```

```
--- Top 5 users by message count ---
1. 44 - 10000 messages
2. 25 - 10000 messages
3. 32 - 10000 messages
4. 41 - 10000 messages
5. 39 - 10000 messages
```

```
--- Top 5 rooms by message count ---
1. 2 - 25000 messages
2. 8 - 25000 messages
3. 7 - 25000 messages
4. 4 - 25000 messages
5. 13 - 25000 messages
```

```
--- Top 5 users by room count ---
1. 8 - 20 rooms
2. 12 - 20 rooms
3. 10 - 20 rooms
4. 7 - 20 rooms
5. 49 - 20 rooms
```

```
=====
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```

## Stress test (1M messages)

client-part2/main.go

Parameter Value

PoolSize 128

UserCount 50

MessageCount 1,000

RoomCount 20

MessageBuffer 120,000

Total messages 1,000,000

consumer-v3.env on EC2

CONSUMER\_WORKERS 5

DB\_WORKERS 3

STATS\_WORKERS 3

BATCH\_SIZE 5000

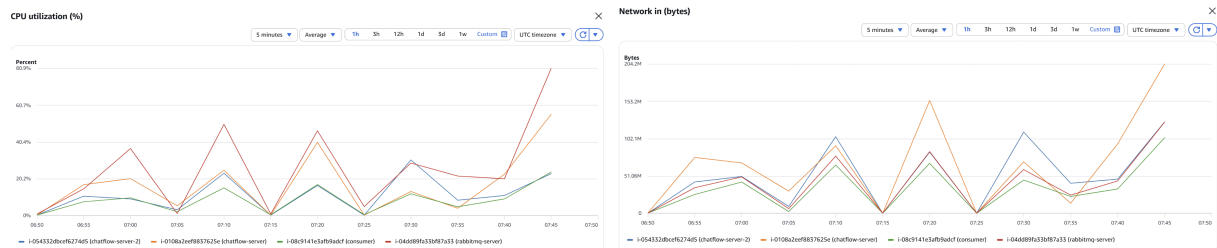
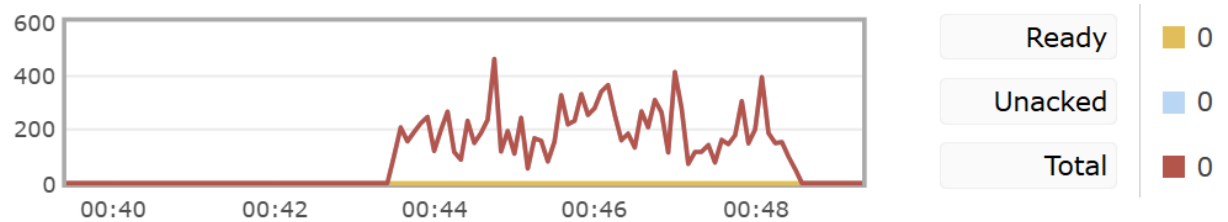
FLUSH\_INTERVAL 1000ms

chat-server .env on EC2

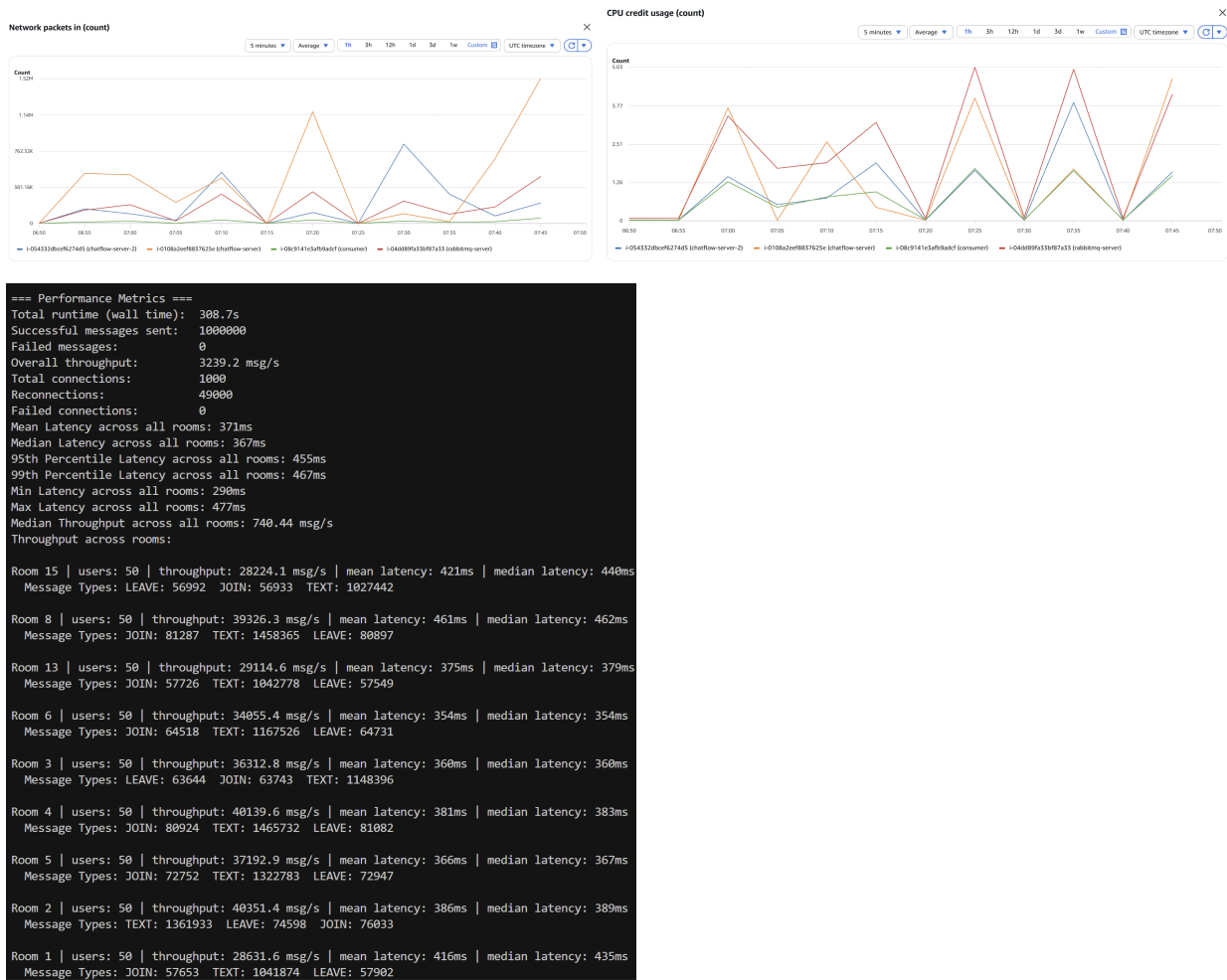
PUBLISH\_WORKERS 20

ROOM\_COUNT 20

Queued messages last ten minutes ?



Sai Supriya Kottru



Sai Supriya Kottru

```
Room 1 | users: 50 | throughput: 28631.6 msg/s | mean latency: 416ms | median latency: 435ms
Message Types: JOIN: 57653 TEXT: 1041874 LEAVE: 57902

Room 14 | users: 50 | throughput: 36220.2 msg/s | mean latency: 339ms | median latency: 338ms
Message Types: JOIN: 70114 TEXT: 1268871 LEAVE: 70200

Room 16 | users: 50 | throughput: 38547.8 msg/s | mean latency: 368ms | median latency: 368ms
Message Types: JOIN: 74287 TEXT: 1340316 LEAVE: 74199

Room 19 | users: 50 | throughput: 36369.2 msg/s | mean latency: 344ms | median latency: 345ms
Message Types: JOIN: 69793 TEXT: 1257643 LEAVE: 69882

Room 7 | users: 50 | throughput: 35618.0 msg/s | mean latency: 368ms | median latency: 368ms
Message Types: JOIN: 70651 TEXT: 1278521 LEAVE: 70454

Room 12 | users: 50 | throughput: 30716.4 msg/s | mean latency: 365ms | median latency: 367ms
Message Types: LEAVE: 58922 JOIN: 58722 TEXT: 1066441

Room 18 | users: 50 | throughput: 44227.2 msg/s | mean latency: 328ms | median latency: 330ms
Message Types: JOIN: 63829 TEXT: 1150794 LEAVE: 63940

Room 20 | users: 50 | throughput: 41020.3 msg/s | mean latency: 385ms | median latency: 383ms
Message Types: JOIN: 75822 TEXT: 1367677 LEAVE: 74823

Room 10 | users: 50 | throughput: 23482.7 msg/s | mean latency: 364ms | median latency: 364ms
Message Types: LEAVE: 44788 JOIN: 44943 TEXT: 808774

Room 11 | users: 50 | throughput: 33276.6 msg/s | mean latency: 367ms | median latency: 368ms
Message Types: LEAVE: 66267 JOIN: 66057 TEXT: 1192033

Room 9 | users: 50 | throughput: 38885.7 msg/s | mean latency: 332ms | median latency: 332ms
Message Types: JOIN: 73639 TEXT: 1325324 LEAVE: 73676

Room 17 | users: 50 | throughput: 32953.7 msg/s | mean latency: 340ms | median latency: 342ms
Message Types: TEXT: 1137153 LEAVE: 63102 JOIN: 62884
```

```
===== Consumer Metrics =====

--- Core ---
Active users (last 1h): 50
Busiest room:          4 (50000 messages)
Most active user:      36 (20 rooms)
  room 18              last active: 2026-03-25T07:48:21.45333Z
  room 3               last active: 2026-03-25T07:48:20.155469Z
  room 6              last active: 2026-03-25T07:48:07.421144Z
  room 5              last active: 2026-03-25T07:47:50.382615Z
  room 12             last active: 2026-03-25T07:47:35.062916Z
  room 7              last active: 2026-03-25T07:47:14.181955Z
  room 19             last active: 2026-03-25T07:47:11.108037Z
  room 17             last active: 2026-03-25T07:46:42.226176Z
  room 16             last active: 2026-03-25T07:46:34.510947Z
  room 13             last active: 2026-03-25T07:46:23.529521Z
  room 9              last active: 2026-03-25T07:45:54.707584Z
  room 14             last active: 2026-03-25T07:45:52.70364Z
  room 1              last active: 2026-03-25T07:45:22.347025Z
  room 8              last active: 2026-03-25T07:45:15.84136Z
  room 15             last active: 2026-03-25T07:45:14.713857Z
  room 11             last active: 2026-03-25T07:44:40.510785Z
  room 4              last active: 2026-03-25T07:44:30.73523Z
  room 20             last active: 2026-03-25T07:43:54.892818Z
  room 10             last active: 2026-03-25T07:43:53.700147Z
  room 2              last active: 2026-03-25T07:43:50.66446Z
```

```
-- Messages/min (last 15 buckets) ---
2026-03-25T07:48:00Z 67713 msg
2026-03-25T07:47:00Z 196327 msg
2026-03-25T07:46:00Z 199670 msg
2026-03-25T07:45:00Z 191040 msg
2026-03-25T07:44:00Z 192768 msg
2026-03-25T07:43:00Z 152482 msg

-- Top 5 users by message count ---
1. 13 - 20000 messages
2. 8 - 20000 messages
3. 1 - 20000 messages
4. 39 - 20000 messages
5. 36 - 20000 messages

-- Top 5 rooms by message count ---
1. 13 - 50000 messages
2. 12 - 50000 messages
3. 14 - 50000 messages
4. 6 - 50000 messages
5. 4 - 50000 messages

-- Top 5 users by room count ---
2. 8 - 20000 messages
3. 1 - 20000 messages
4. 39 - 20000 messages
5. 36 - 20000 messages
```

```
--- Top 5 users by room count ---
2. 8 - 20000 messages
3. 1 - 20000 messages
4. 39 - 20000 messages
5. 36 - 20000 messages

--- Top 5 rooms by message count ---
1. 13 - 50000 messages
2. 12 - 50000 messages
2. 8 - 20000 messages
3. 1 - 20000 messages
4. 39 - 20000 messages
5. 36 - 20000 messages

2. 8 - 20000 messages
3. 1 - 20000 messages
2. 8 - 20000 messages
2. 8 - 20000 messages
2. 8 - 20000 messages
2. 8 - 20000 messages
3. 1 - 20000 messages
4. 39 - 20000 messages
5. 36 - 20000 messages

--- Top 5 rooms by message count ---
1. 13 - 50000 messages
2. 12 - 50000 messages
3. 14 - 50000 messages
2. 8 - 20000 messages
3. 1 - 20000 messages
4. 39 - 20000 messages
5. 36 - 20000 messages
```

```
--- Top 5 rooms by message count ---
1. 13 - 50000 messages
2. 12 - 50000 messages
3. 14 - 50000 messages

--- Top 5 rooms by message count ---
1. 13 - 50000 messages
2. 12 - 50000 messages
3. 14 - 50000 messages
4. 6 - 50000 messages
5. 4 - 50000 messages

--- Top 5 users by room count ---
--- Top 5 users by room count ---
1. 8 - 20 rooms
2. 12 - 20 rooms
3. 10 - 20 rooms
4. 7 - 20 rooms
5. 49 - 20 rooms
```

```
=====
➔ chat-flow git:(assignment-3) 🌸 🍄 🎈
```

Endurance test (30min 4.68M messages)

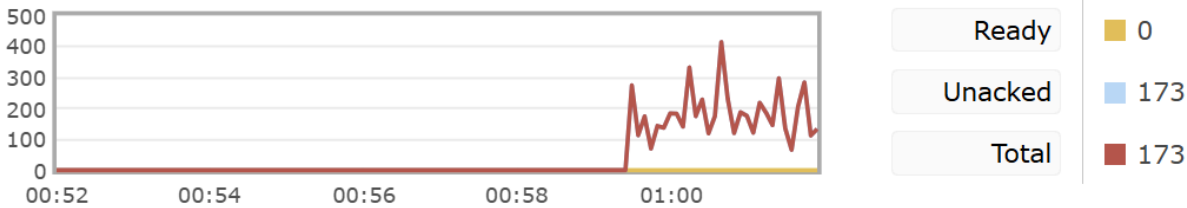
Global counts ?

Connections: 5Channels: 0Exchanges: 8Queues: 41Consumers: 45

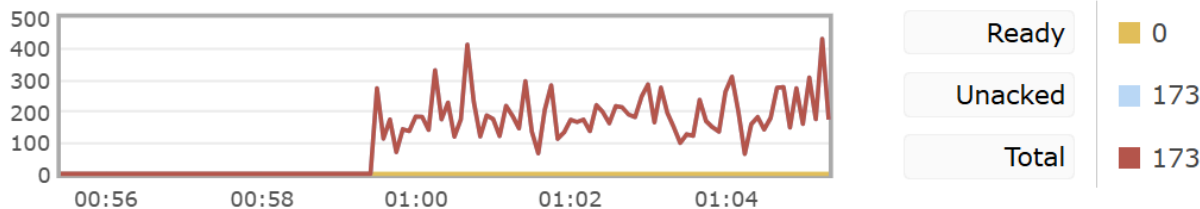
▼ Nodes

| Name                    | File descriptors ?    | Erlang processes         | Memory ?                          | Disk space                      | Uptime | Cores | Info        | Reset stats        | +/- |
|-------------------------|-----------------------|--------------------------|-----------------------------------|---------------------------------|--------|-------|-------------|--------------------|-----|
| rabbit@ip-172-31-25-198 | 65<br>32768 available | 518<br>1048576 available | 134 MiB<br>547 MiB high watermark | 5.6 GiB<br>48 MiB low watermark | 2h 12m | 2     | basic 2 rss | This nodeAll nodes |     |

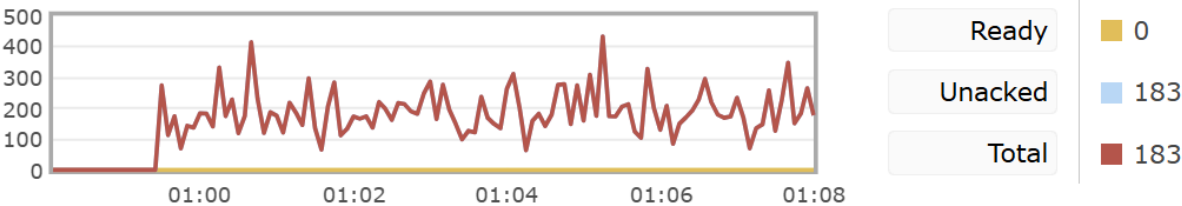
Queued messages last ten minutes ?



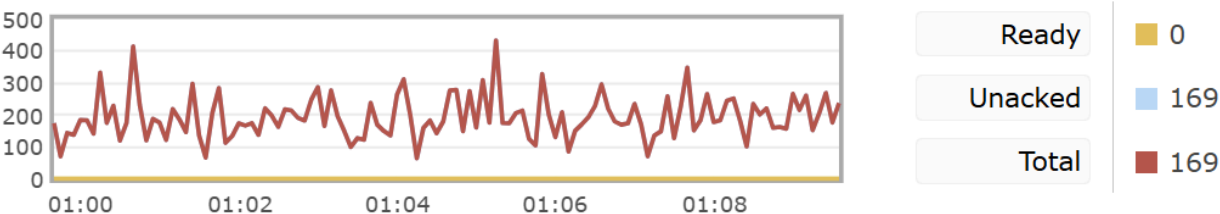
Queued messages last ten minutes ?



Queued messages last ten minutes ?

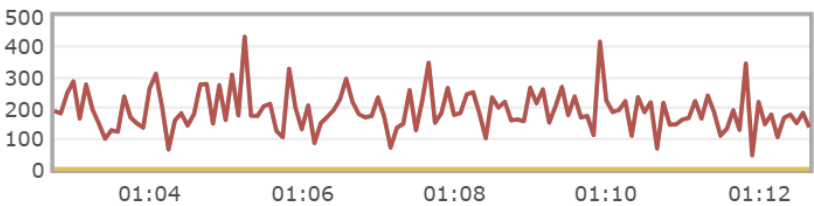


Queued messages last ten minutes ?



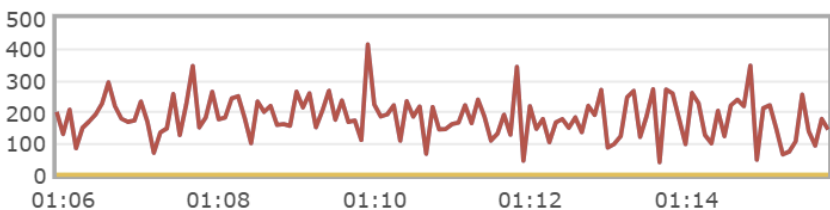


Queued messages last ten minutes ?



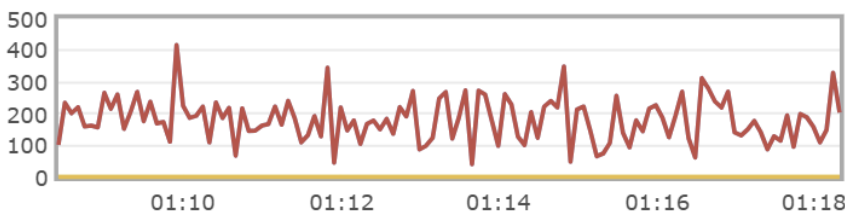
|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 220 |
| Total   | 220 |

Queued messages last ten minutes ?



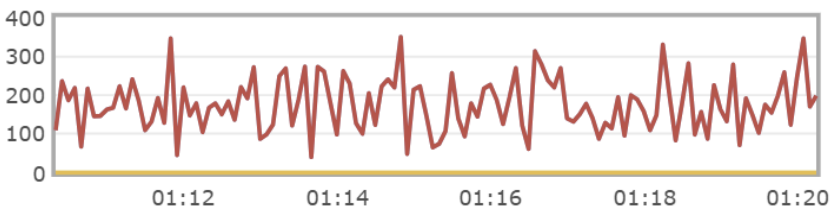
|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 216 |
| Total   | 216 |

Queued messages last ten minutes ?



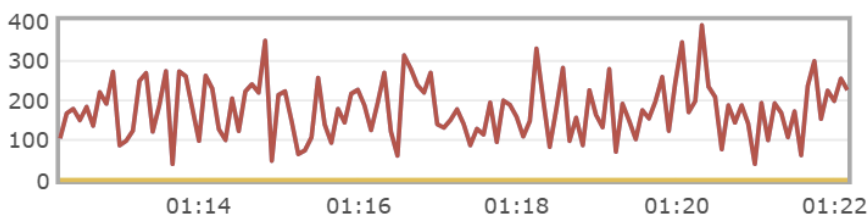
|         |    |
|---------|----|
| Ready   | 0  |
| Unacked | 83 |
| Total   | 83 |

Queued messages last ten minutes ?



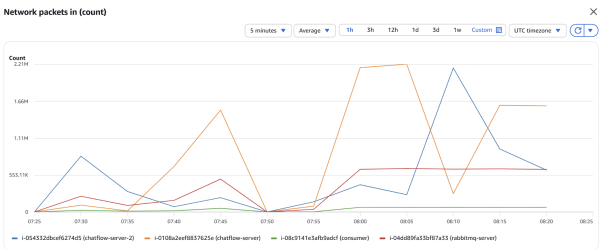
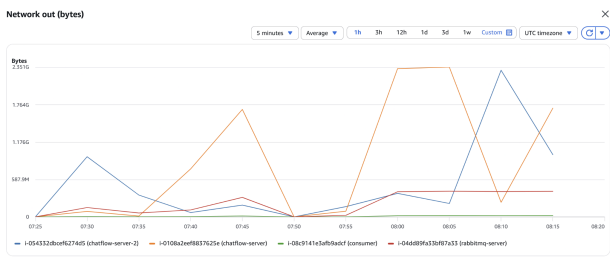
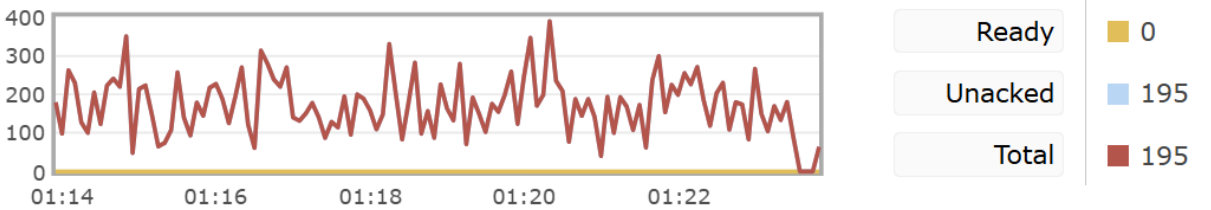
|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 388 |
| Total   | 388 |

Queued messages last ten minutes ?



|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 270 |
| Total   | 270 |

Queued messages last ten minutes ?



# Sai Supriya Kottru

```
=== Performance Metrics ===
Total runtime (wall time): 1450.2s
Successful messages sent: 4680000
Failed messages: 0
Overall throughput: 3227.1 msg/s
Total connections: 1000
Reconnections: 233000
Failed connections: 0
Mean Latency across all rooms: 376ms
Median Latency across all rooms: 385ms
95th Percentile Latency across all rooms: 418ms
99th Percentile Latency across all rooms: 425ms
Min Latency across all rooms: 256ms
Max Latency across all rooms: 437ms
Median Throughput across all rooms: 767.49 msg/s
Throughput across rooms:

Room 4 | users: 50 | throughput: 30191.5 msg/s | mean latency: 372ms | median latency: 372ms
Message Types: JOIN: 278512 TEXT: 5044592 LEAVE: 279137

Room 10 | users: 50 | throughput: 40344.3 msg/s | mean latency: 370ms | median latency: 370ms
Message Types: TEXT: 6701851 LEAVE: 369674 JOIN: 369221

Room 15 | users: 50 | throughput: 39910.4 msg/s | mean latency: 398ms | median latency: 398ms
Message Types: TEXT: 6657343 LEAVE: 368724 JOIN: 369103

Room 9 | users: 50 | throughput: 27227.8 msg/s | mean latency: 382ms | median latency: 384ms
Message Types: TEXT: 4636247 LEAVE: 257347 JOIN: 256459

Room 14 | users: 50 | throughput: 33414.2 msg/s | mean latency: 389ms | median latency: 389ms
Message Types: LEAVE: 313740 JOIN: 313634 TEXT: 5660534

Room 2 | users: 50 | throughput: 30891.8 msg/s | mean latency: 361ms | median latency: 372ms
Message Types: TEXT: 4915664 LEAVE: 272370 JOIN: 272157

Room 19 | users: 50 | throughput: 39195.9 msg/s | mean latency: 417ms | median latency: 417ms
Message Types: LEAVE: 372308 JOIN: 374857 TEXT: 6745469

Room 5 | users: 50 | throughput: 24726.3 msg/s | mean latency: 374ms | median latency: 368ms
Message Types: LEAVE: 229668 JOIN: 229469 TEXT: 4138999

Room 13 | users: 50 | throughput: 35030.3 msg/s | mean latency: 397ms | median latency: 400ms
Message Types: JOIN: 324081 TEXT: 5857363 LEAVE: 323962
```

```
Room 1 | users: 50 | throughput: 28703.9 msg/s | mean latency: 384ms | median latency: 384ms
Message Types: LEAVE: 266440 JOIN: 265553 TEXT: 4807018

Room 3 | users: 50 | throughput: 40059.8 msg/s | mean latency: 387ms | median latency: 387ms
Message Types: LEAVE: 371633 JOIN: 370421 TEXT: 6717716

Room 18 | users: 50 | throughput: 39273.3 msg/s | mean latency: 382ms | median latency: 382ms
Message Types: LEAVE: 368128 JOIN: 367416 TEXT: 6633135

Room 8 | users: 50 | throughput: 39387.0 msg/s | mean latency: 419ms | median latency: 419ms
Message Types: JOIN: 375079 TEXT: 6773568 LEAVE: 373702

Room 20 | users: 50 | throughput: 26243.9 msg/s | mean latency: 374ms | median latency: 375ms
Message Types: LEAVE: 240783 JOIN: 240269 TEXT: 4351723

Room 7 | users: 50 | throughput: 38515.3 msg/s | mean latency: 380ms | median latency: 381ms
Message Types: TEXT: 6415536 LEAVE: 355320 JOIN: 354910

Room 11 | users: 50 | throughput: 37352.5 msg/s | mean latency: 293ms | median latency: 285ms
Message Types: JOIN: 275134 TEXT: 4986112 LEAVE: 276020

Room 12 | users: 50 | throughput: 21571.5 msg/s | mean latency: 387ms | median latency: 387ms
Message Types: TEXT: 3621995 LEAVE: 201304 JOIN: 201088

Room 6 | users: 50 | throughput: 37142.1 msg/s | mean latency: 391ms | median latency: 394ms
Message Types: TEXT: 6164328 LEAVE: 341121 JOIN: 340760

Room 16 | users: 50 | throughput: 37560.9 msg/s | mean latency: 392ms | median latency: 395ms
Message Types: TEXT: 6331455 LEAVE: 351100 JOIN: 350784

Room 17 | users: 50 | throughput: 47195.9 msg/s | mean latency: 264ms | median latency: 264ms
Message Types: JOIN: 318520 TEXT: 5751016 LEAVE: 319134
```

```
===== Consumer Metrics =====

--- Core ---
Active users (last 1h): 50
Busiest room: 17 (284000 messages)
Most active user: 4 (20 rooms)
room 17 last active: 2026-03-25T08:23:28.085183Z
room 11 last active: 2026-03-25T08:23:25.194231Z
room 2 last active: 2026-03-25T08:23:14.238484Z
room 7 last active: 2026-03-25T08:21:11.935457Z
room 20 last active: 2026-03-25T08:20:43.878415Z
room 18 last active: 2026-03-25T08:18:30.137515Z
room 14 last active: 2026-03-25T08:17:56.093648Z
room 9 last active: 2026-03-25T08:17:35.29404Z
room 3 last active: 2026-03-25T08:15:18.855277Z
room 1 last active: 2026-03-25T08:14:35.770848Z
room 15 last active: 2026-03-25T08:12:15.308403Z
room 13 last active: 2026-03-25T08:11:57.266008Z
room 5 last active: 2026-03-25T08:11:44.919149Z
room 10 last active: 2026-03-25T08:09:03.315059Z
room 4 last active: 2026-03-25T08:08:29.480962Z
room 16 last active: 2026-03-25T08:07:59.870451Z
room 6 last active: 2026-03-25T08:05:39.504383Z
room 12 last active: 2026-03-25T08:05:30.663777Z
room 8 last active: 2026-03-25T08:02:25.078869Z
room 19 last active: 2026-03-25T08:02:19.220326Z
```

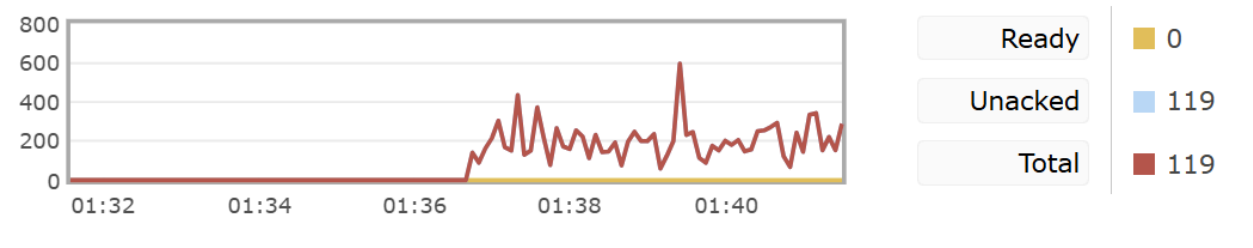
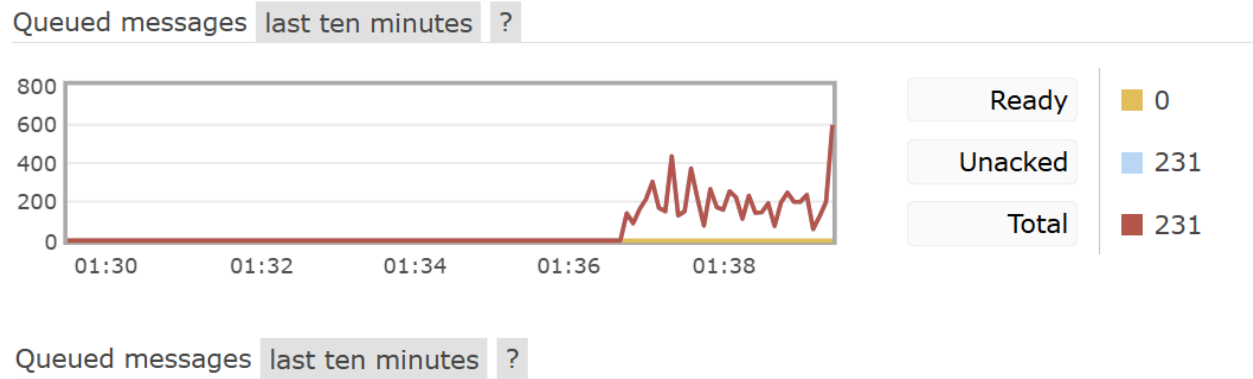
```
--- Messages/min (last 15 buckets) ---
2026-03-25T08:23:00Z 95620 msg
2026-03-25T08:22:00Z 207213 msg
2026-03-25T08:21:00Z 198170 msg
2026-03-25T08:20:00Z 191846 msg
2026-03-25T08:19:00Z 198863 msg
2026-03-25T08:18:00Z 194235 msg
2026-03-25T08:17:00Z 191817 msg
2026-03-25T08:16:00Z 196804 msg
2026-03-25T08:15:00Z 188347 msg
2026-03-25T08:14:00Z 188634 msg
2026-03-25T08:13:00Z 195323 msg
2026-03-25T08:12:00Z 191022 msg
2026-03-25T08:11:00Z 193633 msg
2026-03-25T08:10:00Z 191282 msg
2026-03-25T08:09:00Z 196480 msg
```

```
--- Top 5 users by message count ---
1. 7 - 113593 messages
2. 4 - 113593 messages
3. 22 - 113592 messages
4. 33 - 113592 messages
5. 29 - 113592 messages

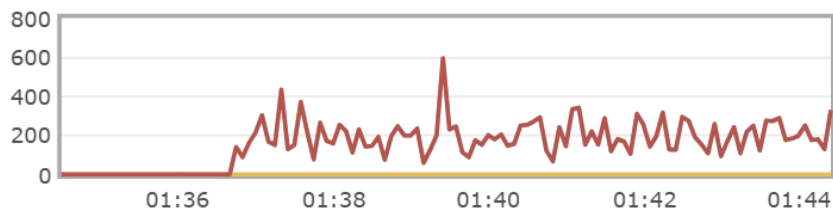
--- Top 5 rooms by message count ---
1. 7 - 284000 messages
2. 15 - 284000 messages
3. 5 - 284000 messages
4. 8 - 284000 messages
5. 17 - 284000 messages

--- Top 5 users by room count ---
1. 8 - 20 rooms
2. 12 - 20 rooms
3. 10 - 20 rooms
4. 7 - 20 rooms
5. 49 - 20 rooms
```

Double Endurance test (48min 9.36M messages)

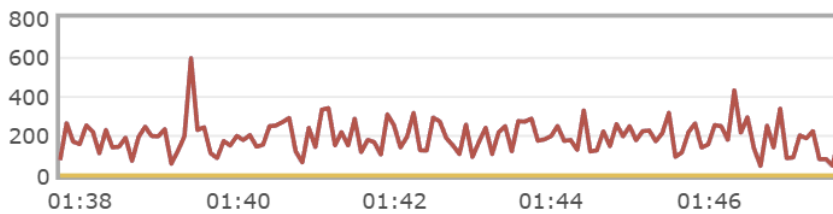


Queued messages last ten minutes ?

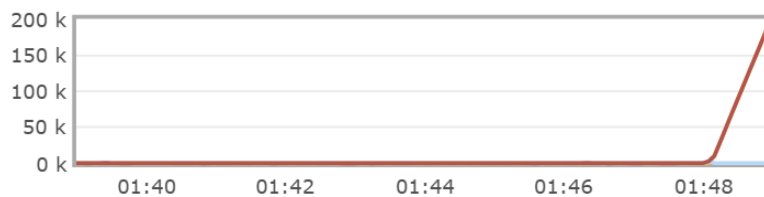


|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 122 |
| Total   | 122 |

Queued messages last ten minutes ?



|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 132 |
| Total   | 132 |



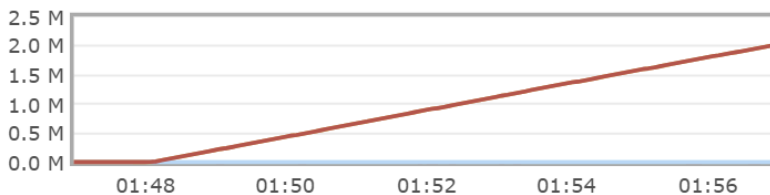
|         |         |
|---------|---------|
| Ready   | 205,214 |
| Unacked | 171     |
| Total   | 205,385 |

Queued messages last ten minutes ?



|         |         |
|---------|---------|
| Ready   | 983,638 |
| Unacked | 223     |
| Total   | 983,861 |

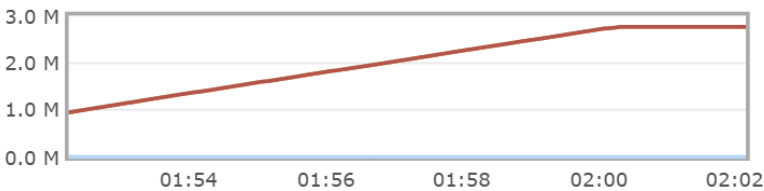
Queued messages last ten minutes ?



|         |           |
|---------|-----------|
| Ready   | 2,020,988 |
| Unacked | 438       |
| Total   | 2,021,426 |

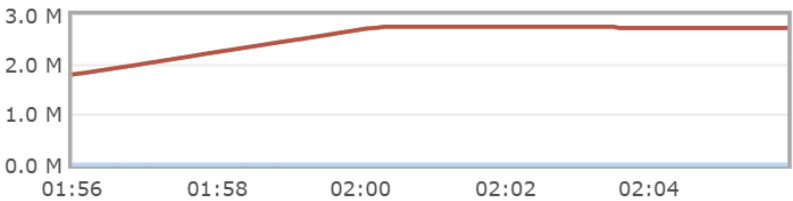
Force stopped the consumer, since the db\_chan was full because DB writes are slower than incoming messages. The consumer goroutines are blocked trying to send to db\_chan, so they're not calling Accept(), which means RabbitMQ credits aren't flowing back, causing the queue to build up.

Queued messages last ten minutes ?



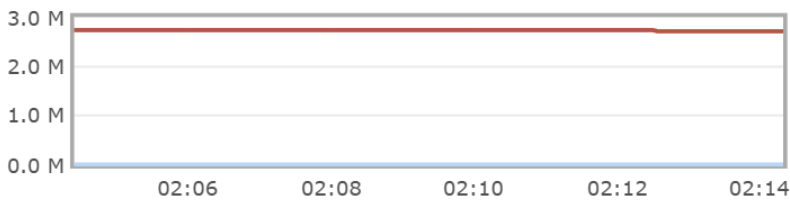
|         |           |
|---------|-----------|
| Ready   | 2,762,212 |
| Unacked | 0         |
| Total   | 2,762,212 |

Queued messages last ten minutes ?



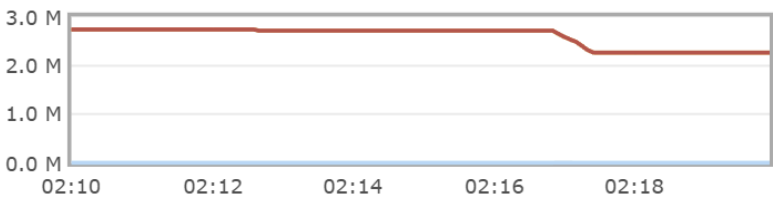
|         |           |
|---------|-----------|
| Ready   | 2,736,347 |
| Unacked | 0         |
| Total   | 2,736,347 |

Queued messages last ten minutes ?



|         |           |
|---------|-----------|
| Ready   | 2,711,306 |
| Unacked | 0         |
| Total   | 2,711,306 |

Queued messages last ten minutes ?

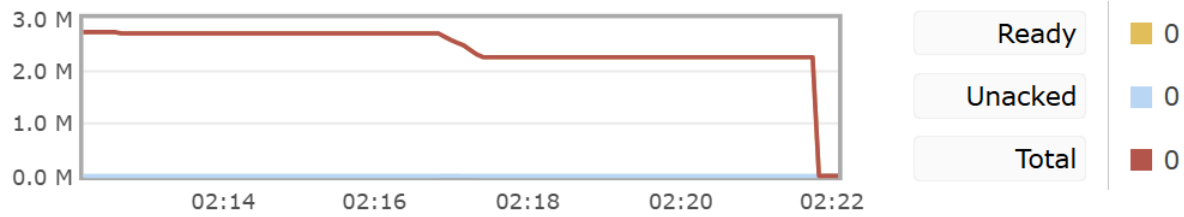


|         |           |
|---------|-----------|
| Ready   | 2,260,536 |
| Unacked | 0         |
| Total   | 2,260,536 |

Sai Supriya Kottru

Deleted the persistence queue

Queued messages **last ten minutes** ?



Message rates **last ten minutes** ?

Waiting for data...

Global counts ?

Connections: **5**

Channels: **0**

Exchanges: **8**

Queues: **40**

Consumers: **40**



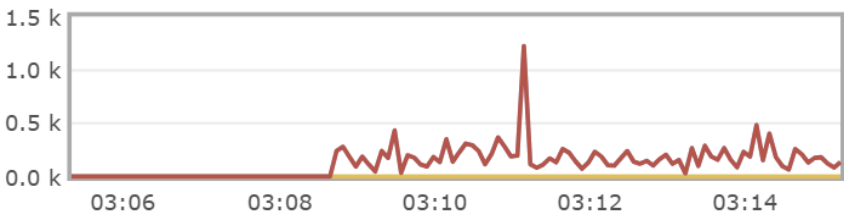
Redo with fix - move Accept() before sending the messages to the consumer

Queued messages last ten minutes ?



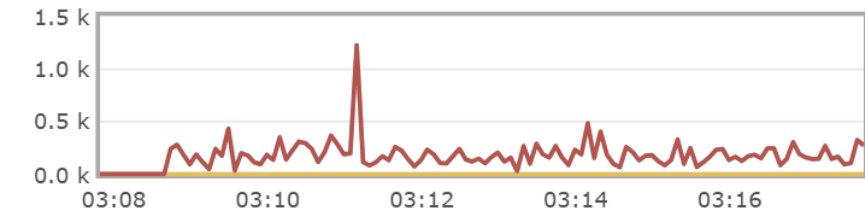
|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 106 |
| Total   | 106 |

Queued messages last ten minutes ?



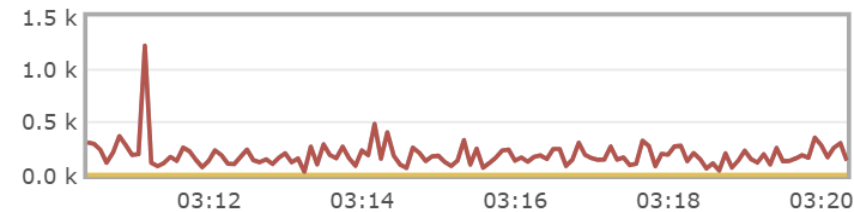
|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 329 |
| Total   | 329 |

Queued messages last ten minutes ?



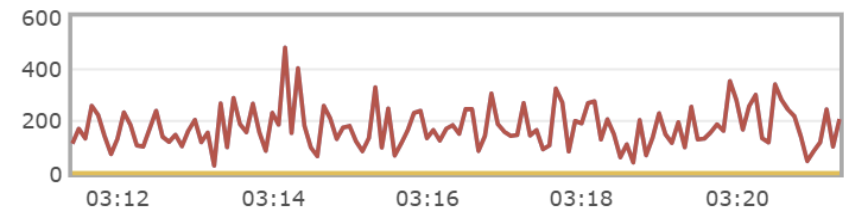
|         |    |
|---------|----|
| Ready   | 0  |
| Unacked | 83 |
| Total   | 83 |

Queued messages last ten minutes ?



|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 118 |
| Total   | 118 |

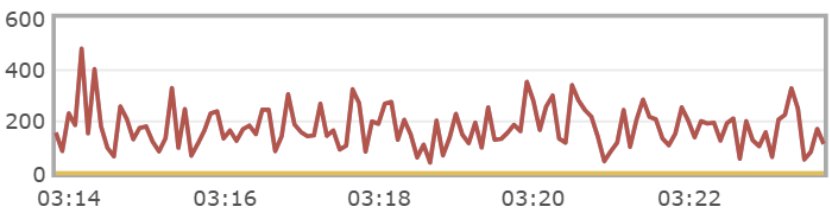
Queued messages last ten minutes ?



|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 284 |
| Total   | 284 |

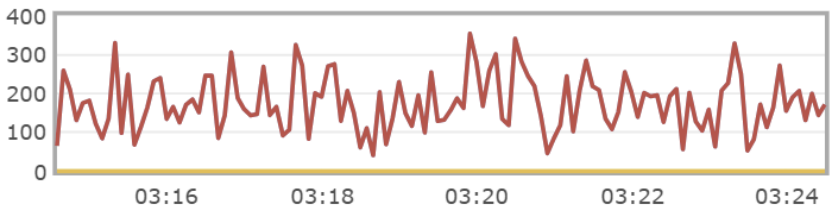


Queued messages last ten minutes ?



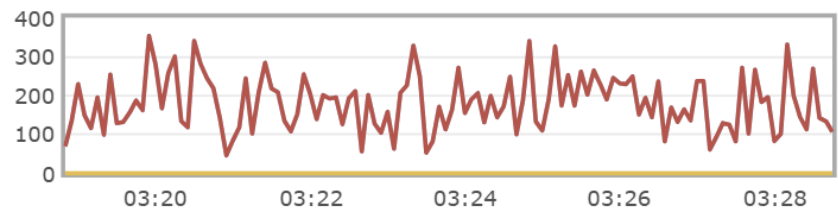
|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 163 |
| Total   | 163 |

Queued messages last ten minutes ?



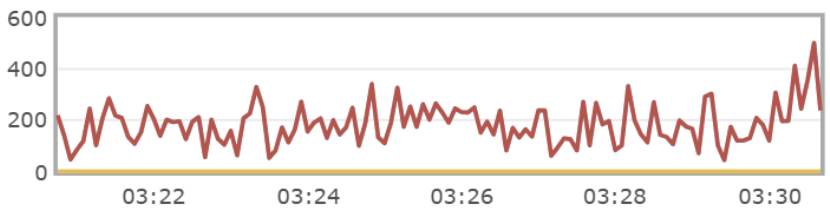
|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 248 |
| Total   | 248 |

Queued messages last ten minutes ?



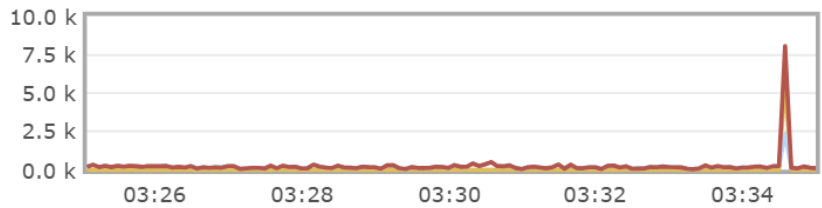
|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 198 |
| Total   | 198 |

Queued messages last ten minutes ?



|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 228 |
| Total   | 228 |

Queued messages last ten minutes ?



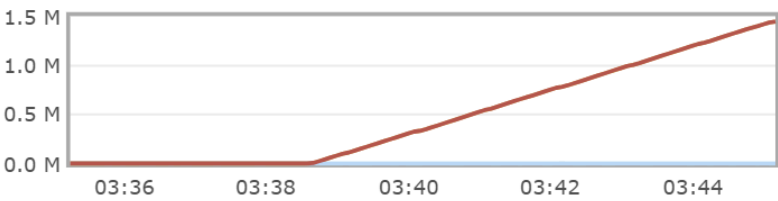
|         |     |
|---------|-----|
| Ready   | 0   |
| Unacked | 170 |
| Total   | 170 |

Queued messages last ten minutes ?



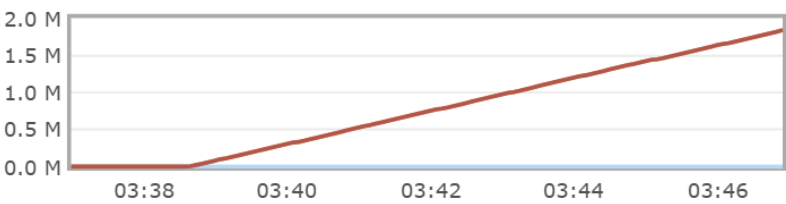
|         |         |
|---------|---------|
| Ready   | 835,279 |
| Unacked | 102     |
| Total   | 835,381 |

Queued messages last ten minutes ?



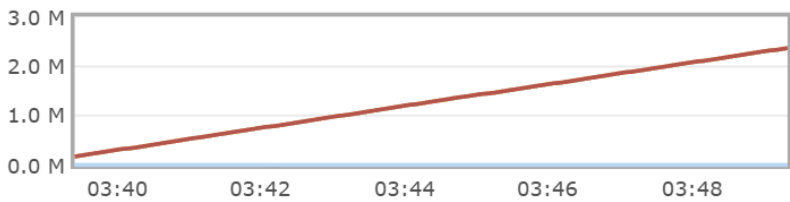
|         |           |
|---------|-----------|
| Ready   | 1,460,631 |
| Unacked | 128       |
| Total   | 1,460,759 |

Queued messages last ten minutes ?



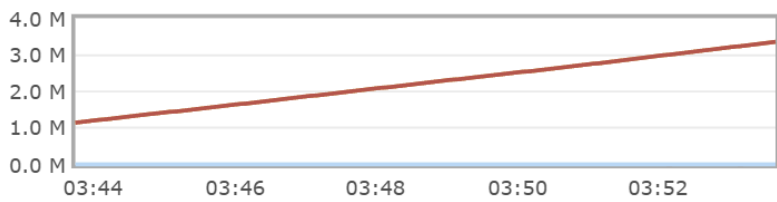
|         |           |
|---------|-----------|
| Ready   | 1,855,998 |
| Unacked | 140       |
| Total   | 1,856,138 |

Queued messages last ten minutes ?



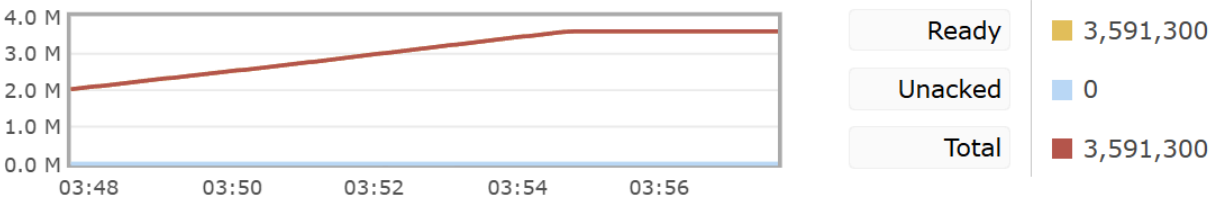
|         |           |
|---------|-----------|
| Ready   | 2,382,962 |
| Unacked | 74        |
| Total   | 2,383,036 |

Queued messages last ten minutes ?

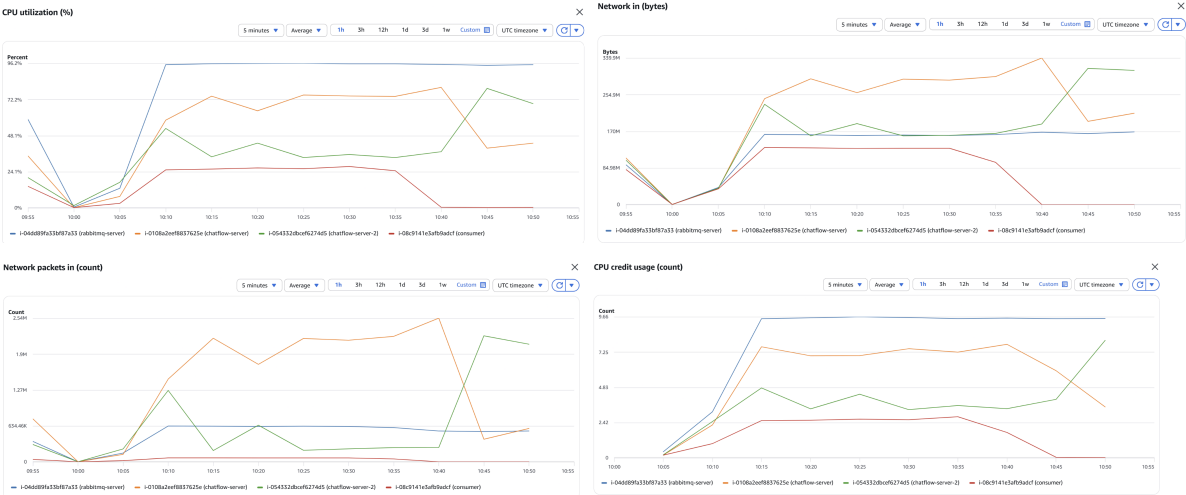


|         |           |
|---------|-----------|
| Ready   | 3,372,062 |
| Unacked | 139       |
| Total   | 3,372,201 |

Queued messages last ten minutes ?



| Overview     |                                             |         |           |       | Messages  |         |           | Message rates |         |     |     | +/- |
|--------------|---------------------------------------------|---------|-----------|-------|-----------|---------|-----------|---------------|---------|-----|-----|-----|
| Virtual host | Name                                        | Type    | Features  | State | Ready     | Unacked | Total     | incoming      | deliver | get | ack |     |
| /            | persistence_queue                           | classic | D Args    | idle  | 3,591,300 | 0       | 3,591,300 |               |         |     |     |     |
| /            | room.1.37ad692d-9af4-46cd-b5e6-a0f031baade4 | classic | D AD Args | idle  | 0         | 0       | 0         |               |         |     |     |     |
| /            | room.1.6f69f1e0-6218-4497-8fad-c075f925c9d7 | classic | D AD Args | idle  | 0         | 0       | 0         |               |         |     |     |     |



# Sai Supriya Kottru

```
=== Performance Metrics ===
Total runtime (wall time): 2773.9s
Successful messages sent: 936000
Failed messages: 0
Overall throughput: 3374.3 msg/s
Total connections: 1000
Reconnections: 46700
Failed connections: 0
Mean Latency across all rooms: 356ms
Median Latency across all rooms: 372ms
95th Percentile Latency across all rooms: 400ms
99th Percentile Latency across all rooms: 408ms
Min Latency across all rooms: 269ms
Max Latency across all rooms: 415ms
Median Throughput across all rooms: 731.94 msg/s
Throughput across rooms:

Room 18 | users: 50 | throughput: 33065.3 msg/s | mean latency: 283ms | median latency: 275ms
Message Types: JOIN: 432449 TEXT: 7804608 LEAVE: 432333

Room 3 | users: 50 | throughput: 39300.8 msg/s | mean latency: 400ms | median latency: 400ms
Message Types: TEXT: 12981464 LEAVE: 712557 JOIN: 719359

Room 12 | users: 50 | throughput: 40541.5 msg/s | mean latency: 379ms | median latency: 380ms
Message Types: TEXT: 13658827 LEAVE: 757438 JOIN: 755708

Room 20 | users: 50 | throughput: 22842.9 msg/s | mean latency: 376ms | median latency: 377ms
Message Types: JOIN: 425973 TEXT: 7689096 LEAVE: 426971

Room 9 | users: 50 | throughput: 33403.4 msg/s | mean latency: 376ms | median latency: 377ms
Message Types: JOIN: 628261 TEXT: 11330645 LEAVE: 628488

Room 1 | users: 50 | throughput: 36120.8 msg/s | mean latency: 377ms | median latency: 376ms
Message Types: LEAVE: 677003 JOIN: 676812 TEXT: 12219371

Room 13 | users: 50 | throughput: 40614.4 msg/s | mean latency: 341ms | median latency: 341ms
Message Types: JOIN: 661483 TEXT: 11965914 LEAVE: 661886

Room 8 | users: 50 | throughput: 46268.9 msg/s | mean latency: 277ms | median latency: 278ms
Message Types: JOIN: 578449 TEXT: 10428273 LEAVE: 576560
```

```
Room 4 | users: 50 | throughput: 39149.8 msg/s | mean latency: 400ms | median latency: 400ms
Message Types: LEAVE: 715315 JOIN: 722888 TEXT: 13035164

Room 15 | users: 50 | throughput: 21120.7 msg/s | mean latency: 372ms | median latency: 373ms
Message Types: JOIN: 389083 TEXT: 7026201 LEAVE: 388876

Room 5 | users: 50 | throughput: 41215.0 msg/s | mean latency: 376ms | median latency: 376ms
Message Types: JOIN: 750777 LEAVE: 751864 TEXT: 13556817

Room 16 | users: 50 | throughput: 31756.1 msg/s | mean latency: 331ms | median latency: 328ms
Message Types: JOIN: 511312 TEXT: 9244208 LEAVE: 512645

Room 19 | users: 50 | throughput: 28649.0 msg/s | mean latency: 320ms | median latency: 328ms
Message Types: LEAVE: 446496 JOIN: 444048 TEXT: 8065621

Room 2 | users: 50 | throughput: 25919.7 msg/s | mean latency: 382ms | median latency: 385ms
Message Types: JOIN: 490508 TEXT: 8837544 LEAVE: 491405

Room 11 | users: 50 | throughput: 25525.0 msg/s | mean latency: 372ms | median latency: 373ms
Message Types: JOIN: 472578 TEXT: 8528705 LEAVE: 473851

Room 6 | users: 50 | throughput: 25204.2 msg/s | mean latency: 333ms | median latency: 335ms
Message Types: JOIN: 409475 TEXT: 7405811 LEAVE: 410639

Room 17 | users: 50 | throughput: 38134.9 msg/s | mean latency: 376ms | median latency: 377ms
Message Types: TEXT: 12957620 LEAVE: 718489 JOIN: 717285

Room 7 | users: 50 | throughput: 31069.0 msg/s | mean latency: 372ms | median latency: 372ms
Message Types: TEXT: 10436789 LEAVE: 578832 JOIN: 578704

Room 14 | users: 50 | throughput: 24443.7 msg/s | mean latency: 344ms | median latency: 359ms
Message Types: LEAVE: 425331 JOIN: 424295 TEXT: 7662138

Room 10 | users: 50 | throughput: 42239.1 msg/s | mean latency: 340ms | median latency: 341ms
Message Types: JOIN: 693847 TEXT: 12583159 LEAVE: 696200

[metrics] failed to reach http://54.218.54.179:8080/metrics: Get "http://54.218.54.179:8080/metrics": context deadline exceeded (Client.Timeout exceeded while awaiting headers)
```